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Alarm timing, trust and driver expectation for forward collision warning systems

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Abstract

In order to improve road safety, automobile manufacturers are now developing Forward Collision Warning Systems (FCWS). However, there has been insufficient consideration of how drivers may respond to FCWS. This driving simulator study focused on alarm timing and its impact on driver response to alarm. The experimental investigation considered driver perception of alarm timings and its influence on trust at three driving speeds (40, 60 and 70 mile/h) and two time headways (1.7 and 2.2 s). The results showed that alarm effectiveness varied in response to driving conditions. Alarm promptness had a greater influence on ratings of trust than improvements in braking performance enabled by the alarm system. Moreover, alarms which were presented after braking actions had been initiated were viewed as late alarms. It is concluded that drivers typically expect alarms to be presented before they initiate braking actions and when this does not happen driver trust in the system is substantially decreased.

Keywords: Alarm timing; Driving; Trust

1. Introduction

As a result of increasing intolerance it is now widely recognised that road accidents are a social problem that require a significant reduction throughout the world. Advanced Driver Assistance Systems (ADAS) have been developed in order to improve road safety and increase driver comfort and, as a part of that technical development, automobile manufacturers are now producing Forward Collision Warning Systems (FCWS). It has been assumed that FCWS will be of great benefit to drivers who fail to pay sufficient attention to the road ahead, resulting in decreased numbers of accidents (Alm and Nilsson, 2000; Ben-Yaacov et al., 2002). On the other hand, however, it has been argued that optimistic predictions of the promised benefits of technically advanced systems have not taken adequate consideration of human factors which may limit system performance (Kantowitz, 2000; Hancock and Verwey, 1997). Lee and Kantowitz (1998) note that system utility depends not only on the intelligence of the technology but also on the joint performance of FCWS and drivers.

The criteria that determine alarm activation are a critical issue in alarm system design and they are a common human factors discussion topic in FCWS (Janssen and Nilsson, 1993; Hancock and Parasuraman, 1992). As phenomena, rear-end collisions are highly time critical and it is generally assumed that early alarms have the greatest potential to prevent drivers from involvement in collision accidents by giving them more time to respond to imminent collision situations. Lee et al. (2002) found that early alarms led to quicker braking responses by drivers compared with unassisted driving. However, it has been argued that warnings issued at a very early stage in potential collision situations are frequently viewed as false alarms (Janssen et al., 1993; Parasuraman et al., 1997) and these alarms typically have a negative impact on total system performance (Dingus et al., 1998). Bliss and Acton (2003) showed that driver response to alarm systems varied in response to differences in alarm reliability. On the other hand, presenting alarms later may reduce the chances of

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drivers having time to take appropriate action. There is no easy way to overcome the dilemma and it is necessary to accumulate research evidence on how drivers respond to alarms in various driving conditions in order to establish the design of optimum alarm timings.

It is well known that the operator's trust in automated systems plays an important role in determining their interactions with such systems (Lee and Moray, 1992; Muir, 1994; Muir and Moray, 1996; Satchell, 1998) and, in particular, operator trust is strongly linked to reliance on automated systems and patterns of system usage (Parasuraman and Riley, 1997). Lee and Kantowitz (1998) note that trust is one of the most important driver cognitive characteristics which determines the appropriate use of driver support systems. If drivers distrust FCWS, then, in extreme cases, drivers may ignore alarms which are presented from such systems. Hence, driver trust in FCWS requires investigation in order to achieve appropriate levels of cooperation between driver assistance system and driver.

With respect to FCWS, it has been established that the alarm timing may influence driver trust. Abe and Richardson (2004) found that trust in early alarms was higher than that with late alarms under fixed driving conditions. However, Lee and Kantowitz (1998) insist that the design characteristics of ADAS should be considered from three points of view; driver characteristics, system capabilities and environmental factors. This suggests that the relationship between alarm timing and trust may be influenced by environmental factors such as driving speed and time headway. In fact, Green (2000) showed that the level of driving demand resulting from driving speed, time headway, etc. led to variation in braking response. Hence, it is necessary to consider alarm timing and its relation to driver trust in response to various driving conditions to fully understand their inter-relationship.

However, alarm timing not only represents a system capability it also exerts an influence on driver expectation and acceptance of FCWS as a support system. Reichart (1993) notes that the behaviour of ADAS has to be consistent with driver expectation and that a conflict between alarm timing and driver expectation may lead to a reduction in driver trust. This is because alarm timings are viewed as an output which represents system behaviour by drivers. Furthermore, a defining characteristic of FCWS is that they are support systems which have the potential to save drivers in critical situations rather than simply increase driving comfort. Therefore, lack of consistency may seriously decrease system utility and driver expectation should be incorporated into the design of alarm timing. However, there has been little investigation of driver expectation regarding alarm timing and its relation to trust.

The aims of this study were therefore to understand how variation in alarm timing might influence driver behaviour and trust under different driving conditions (three driving speeds and two time headways) and to investigate driver expectations of alarm timing for FCWS and the relationship of this to driver trust.

2. Method

2.1. Apparatus

The research was undertaken using the Leeds Advanced Driving Simulator (2004), which uses complex computer graphics to provide a highly realistic driving environment. The simulated horizontal forward field of view was 230° and the vertical field of view was 39°. A rear view (60° in the horizontal plane) was back projected onto a screen behind the car to provide an image that could be seen through the vehicle's rear view and wing mirrors. The car used in this study was a Rover 216GTi four-door sedan modified for use in the simulator. Although the simulator had a fixed base, physical and functional fidelity were reasonably high and drivers were provided with realistic driving conditions. Drivers received visual and haptic feedback regarding vehicle control from the steering wheel, brakes, accelerator, and gearshift controls in exactly the same way as in a real car. Moreover, computer-generated visual and auditory stimuli accurately represented environmental changes resulting from driver actions. The simulated driving environment was created using Multi-Gen Creator software. In this study, a simulated rural road environment was created with a single carriageway, a leading vehicle and the occasional appearance of oncoming vehicles.

The following vehicle, 'driven' by the subjects, was equipped with a simulated FCWS. Although several algorithms have been used for alarm trigger logics, a Stop-Distance-Algorithm (SDA) was used in this experiment (Wilson et al., 1997). The SDA is so termed because a warning distance is defined based on the difference between the stopping distances of the leading and following vehicle. This algorithm has three free parameters: reaction time (RT), deceleration of the leading vehicle (D_l) and deceleration of the following vehicle (D_f). RT is the assumed reaction time of the driver of the following vehicle and D_l and D_f are the assumed decelerations of the leading and following vehicle, respectively. A warning distance (D_w) is found on the basis of these parameters along with velocities of the leading (V_l) and following (V_f) vehicle as follows:

$$D_w = V_f RT + \frac{V_f^2}{2D_f} - \frac{V_l^2}{2D_l}. \tag{1}$$

In this study, two different timing characteristics of the alarm timing (late/early) were prepared by manipulating D_f (0.60 g for late, 0.45 g for early). The other parameters had the same value in all conditions ($RT = 1.25$ s, $D_l = 0.5$ g). These parameters were chosen by considering two issues. The first was correspondence with 'real-world' values; the values are representative of those that would be experienced in on-road driving and should therefore appear appropriate to the participants. The second was the relative difference in perceived alarm promptness between the alarm timings. It was important that the participants perceived clear differences between the parameter values used. The warning system provided drivers with a simple auditory alarm which lasted approximately 2.0 s and the prominent frequency of the tone was 4000 Hz.

2.2. Subjects and experimental design

Twenty-two participants, ranging in age from 23 to 61 years (M = 31.9, SD = 9.3), took part in the experiment. All were licensed drivers with driving experience of 2 years or more. Each received a payment of £10 for their participation.

The subjects were randomly and evenly assigned to each of the early and late alarm timing conditions. Each condition comprised two sessions; a session without alarms and a session with alarms. Each session included six potential collision events in which the leading vehicle decelerated at a rate of 0.8 g, with the brake lights illuminated, until it came to a stop. Three different driving speeds (40, 60, 70 mile/h) and two different time headways (1.7 and 2.2 s) were implemented as driving conditions. In combination, these gave 6 treatment conditions (1.7 s/40 mile/h; 2.2 s/40 mile/h; 1.7 s/60 mile/h; 2.2 s/60 mile/h; 1.7 s/70 mile/h; 2.2 s/70 mile/h). The order of driving conditions was counterbalanced (Fig. 1) and the order of trials within each session was also counterbalanced across the subjects. The braking events were presented with an irregular time interval to prevent drivers from predicting their occurrence.

In situations where a collision event was engineered, one of the driving conditions discussed above applied. The required conditions were achieved by (1) ensuring that subjects drove at the target speed with a tolerance of ±5% and (2) by adjusting the speed of the leading vehicle to provide the correct headway. The time headway between the leading and following vehicle was fixed by manipulating the leading vehicle's speed in response to the following vehicle's speed. Accordingly the distance headway was determined by the experimental conditions that applied, i.e. the target speed (±5%) and the time headway before the occurrence of a potential collision event.

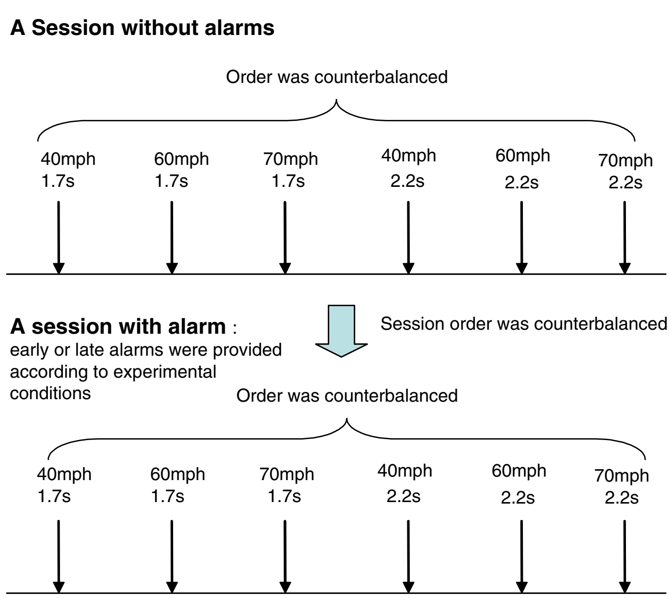

Fig. 1. Driving conditions for each session.

2.3. Procedure

All participants were required to confirm their informed consent and were then briefed on task requirements by the experimenter. Each participant was given a 10-min practice drive to familiarise themselves with the simulator and to experience the lead vehicle decelerations that would be involved in the experimental trials. After a 5-min break, the experimental trials were started. All participants were given two objectives in these trials. One was to keep their own speed to the target speed as directed by the experimenter (40, 60 or 70 mile/h) and the other was to follow the leading vehicle but to avoid rear-end collisions. The drivers were directed to only use the brakes to avoid a collision with the lead vehicle, rather than to change lanes. About 8 s after each braking event, the leading vehicle gradually began to move forward again and the subjects were instructed to gradually accelerate until they reached the target speed. The participants had a 5-min break between sessions.

2.4. Dependent measures

Three dependent measures describing the driver's braking response and two dependent measures describing driver trust in the system and perception of alarm timing were recorded:

(i) *Braking event to brake onset time*: This is the time period between the braking event and application of the brakes. This transition time represents the time period between the occurrence of a critical event and the driver's final action, i.e. implementation of the brakes. The following two dependent variables (ii) and (iii) represent sub-components of braking reaction time.
(ii) *Braking event to accelerator release time*: This is the time period between the braking event and the driver's release of the accelerator.
(iii) *Accelerator release to brake onset time*: This corresponds to the time between the driver's release of the accelerator and application of the brakes.
(iv) *Driver subjective rating of trust*: A 10-point rating scale was used to record the driver's subjective estimation of trust; 1 indicating "not at all" and 10 indicating "completely". After each braking event, the participants gave a verbal judgement of their trust in

response to the following question; *How much do you trust the warning system?*

(v) *Driver subjective perception of alarm timing*: Driver perception of alarm timing was reported by using a simple three-level estimation; 'early', 'correct', or 'late'. All subjects gave a verbal judgement of their perception of alarm timing in response to the following question; *How would you rate the alarm timing?*

3. Results and discussion

In the subsequent statistical analyses an ANOVA was used in order to investigate differences between factors, however, several nonparametric tests were used in response to experimental conditions when homogeneity of variance was not present. None of the participants suffered collisions and so the effect of collision events on driver behaviour could be ignored.

Two different alarm timings (relatively early and relatively late) were achieved by manipulating the alarm trigger parameters. In the following analysis of driver behaviour, *alarm time* (i.e. the elapsed time from when the leading vehicle began to brake to when the alarm was presented) was considered as a contributing factor in determining driver behaviour. Table 1 presents mean alarm times (and standard deviations) for the actual values achieved in the different experimental conditions. It can be clearly seen that vehicle speed has a much smaller influence on alarm time compared with headway value.

3.1. Braking event to brake onset time

Braking event to brake onset time was analysed in order to estimate the effect of the presentation of alarms on net reaction to imminent collision situations.

The first analysis considered the effect of early alarm timing with respect to time headways. Fig. 2 illustrates mean values for braking event to brake onset time in response to time headway. For short time headways, it appears that braking event to brake onset time tends to decrease when an alarm is provided. The standard deviation for the early alarm timing condition is also smaller than that for the no alarm condition. There was a significant difference in variance between 'early alarm timing' and 'no alarms' conditions, $F(1,64) = 8.42$, $p < 0.01$. With respect to the mean values, a signed rank sum test showed that there was a significant difference between the two conditions ($Z = 2.09$, $p < 0.05$).

For long time headways, there was no significant difference between the conditions. However, it appeared that, compared to the early alarm timing condition, the transition time for the no alarm condition varied widely. In fact, there was a significant difference in deviation between the two conditions, $F(1,64) = 4.72$, $p < 0.05$. With respect to interactions between driving speed and the presentation of alarms, no statistically significant interactions were observed.

To sum up, the results suggest that the effect of the presentation of alarms may vary according to driving conditions even when the same alarm trigger logic is used. The explanation for this phenomenon may derive from the timing of the alarm presentation in response to driving conditions. In other words, if alarms are presented at the mean value of alarm time for relatively early alarms in short headway driving (i.e. 0.149 s) then these alarms may decrease braking reaction compared to the no alarm condition. However, if alarms are presented at the mean

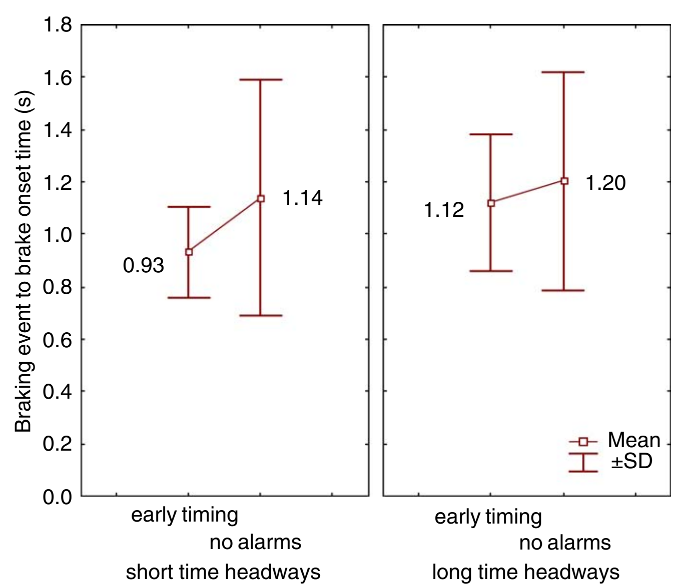

Fig. 2. The impact of the presentation of early alarm timing on braking event to brake onset time in response to time headways.

Table 1
Summary of alarm time for each driving condition

		Velocity		
		40 mile/h	60 mile/h	70 mile/h
Time headway	Long (2.2 s)	Early: 0.641 (0.071) Late: 1.104 (0.126)	Early: 0.533 (0.105) Late: 1.442 (0.262)	Early: 0.498 (0.103) Late: 1.484 (0.439)
	Short (1.7 s)	Early: 0.190 (0.078) Late: 0.661 (0.081)	Early: 0.182 (0.081) Late: 0.782 (0.080)	Early: 0.074 (0.056) Late: 0.859 (0.110)

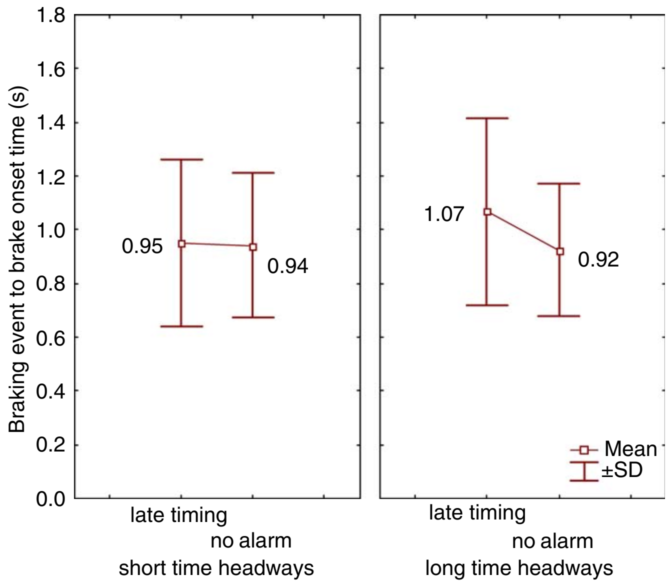

Fig. 3. The impact of the late timing of the alarms on braking event to brake onset time in response to time headways.

value of alarm time for relatively early alarms in long headway driving (i.e. 0.557 s) these alarms have no potential to decrease braking reaction. Furthermore, with respect to the overall effects of alarms on driver behaviour, the presentation of alarms at the mean value of alarm time for relatively early alarms for all driving conditions (i.e. 0.353 s) may lead to more consistent braking reaction to imminent collision situations than when no alarms are provided.

The effect of late alarm timing is considered next. Fig. 3 illustrates mean values of braking event to brake onset time in response to time headway. For short time headways, there is no significant difference between 'late timing' and 'no alarms' conditions. For long time headways, a statistically significant difference in mean values was observed between the two conditions, $F(1,53) = 5.32$, $p < 0.05$. With respect to interactions between driving speed and the presentation of alarms, no statistically significant interactions were observed.

The results suggest that the presentation of late alarms has no potential to improve driver braking reaction time to imminent collision situations, and with longer headways a later alarm timing may actually lead to an increase in braking event to brake onset time.

3.2. *The timing of accelerator release and accelerator release to brake onset*

Changes in the timing of release of the accelerator in response to the timing of the alarms were considered next. Fig. 4 illustrates braking event to accelerator release time for 'early timing' and 'no alarm' conditions in response to time headways. As with short time headways, a signed rank sum test showed that there was a significant difference between the two conditions ($Z = 2.09$, $p < 0.05$). With respect to long time headways, the presentation of alarms did not affect the timing of release of the accelerator. No statistically significant interactions between driving speed and the presentation of alarms were observed.

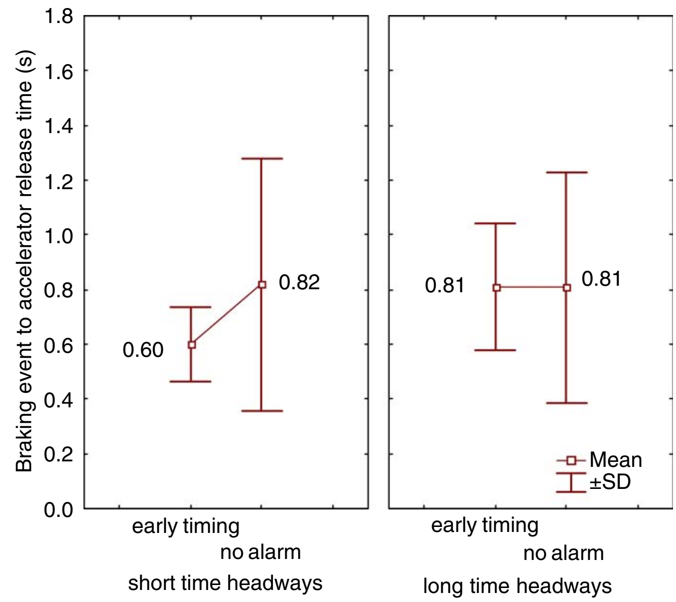

Fig. 4. The impact of the early timing of the alarms on braking event to accelerator release time in response to time headways.

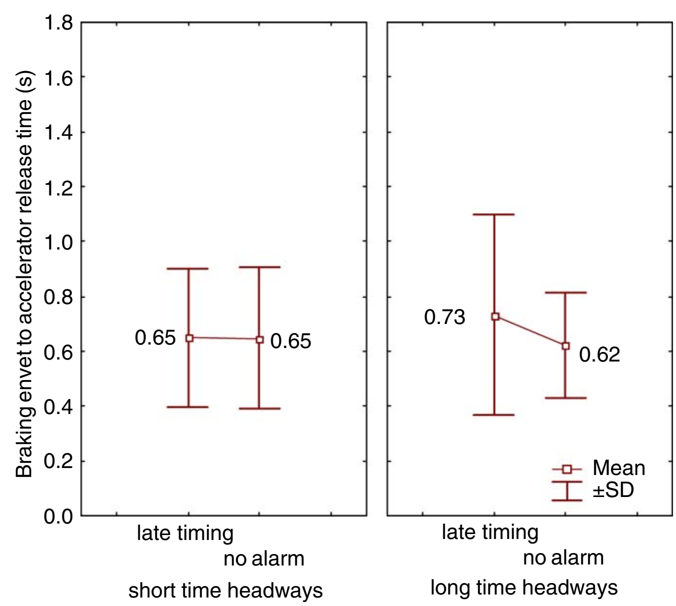

Fig. 5. The impact of the late timing of the alarms on braking event to accelerator release time in response to time headways.

Fig. 5 presents the effect of the presentation of late alarms on the time from braking event to accelerator release in response to time headway. With respect to long time headways it appeared that, compared with the no alarm condition, the transition time for the late timing condition varied widely. A Levene's test revealed that there was a significant difference in variance between the two conditions, $F(1,63) = 8.10$, $p < 0.01$. However, there was no significant difference in mean values between the two conditions. With respect to short time headways, no statistically significant differences were observed. No statistically significant interactions between driving speed and the presentation of alarms were observed.

With respect to accelerator release to brake onset time, no statistically significant differences were observed between the two alarm conditions and time headways. It can be said that the presentation of alarms may have no particular potential to impact on driver braking response after release of the accelerator.

These results are partly consistent with the earlier results related to driver braking response. That is, decreased braking event to brake onset time may derive from prompt release of the accelerator, depending on the timing of the alarms. Furthermore it is possible that the increased variation in accelerator release timing which is associated with late alarms when driving with long time headways, may be linked to delayed driver braking response.

How should driver performance with late alarms when driving with relatively long headways, be interpreted? In relatively low severity car-following situations, for example with long headways, drivers have more time to decide when to start braking compared with short headway driving. Furthermore, for drivers who know alarms will be provided in imminent collision situations, they may expect to respond on the presentation of alarms. Accordingly, due to combined effect of alarm timing and driving demand, late alarms may induce increased variation in accelerator release timing, resulting in longer braking reaction time.

3.3. Driver trust according to driving conditions

Fig. 6 shows the effect of time headways on driver trust for early and late alarms. It appeared that ratings of trust for early alarm timing were similar for both short and long headways. However with respect to late alarm timing, it seemed that trust ratings for long time headways were lower than those for short time headways. A two-way ANOVA found a significant interaction between time headway and alarm timing $F(1, 115) = 4.51$, $p = 0.0357$. A Tukey's post-hoc test revealed a significant difference in mean values for late alarm between two time headways ($p < 0.01$).

The results suggest that trust in early alarms was higher, irrespective of time headways. Furthermore, time headway may affect driver judgement of trust with late alarms.

3.4. Driver perceived alarm timing and its relation to trust and alarm time

The relationship between driver perceived alarm timing and subjective ratings of trust will now be considered. First, the relationship between pre-determined alarm timing (early and late alarm timings) and perceived alarm timing (Table 2) is confirmed. As can be seen in this table, early alarm timing had a tendency to be perceived as correct timing while late alarm timing was perceived as late in the driving conditions used in this study.

Fig. 7 illustrates ratings of trust in alarms which were estimated as having a correct or late timing. There was a significant difference between the two trust values, $F(1.115) = 118.32$, $p < 0.01$. The result suggests that driver perception of alarm timing reflects ratings of trust and, more specifically, trust in alarms which were judged as late was lower than trust in alarms judged as having correct timing.

Table 2
Summary of perceived alarm timing and pre-determined alarm timing

	Early (perceived)	Correct (perceived)	Late (perceived)
Early alarm timing	1 (2%)	43 (65%)	22 (33%)
Late alarm timing	—	18 (34%)	35 (66%)

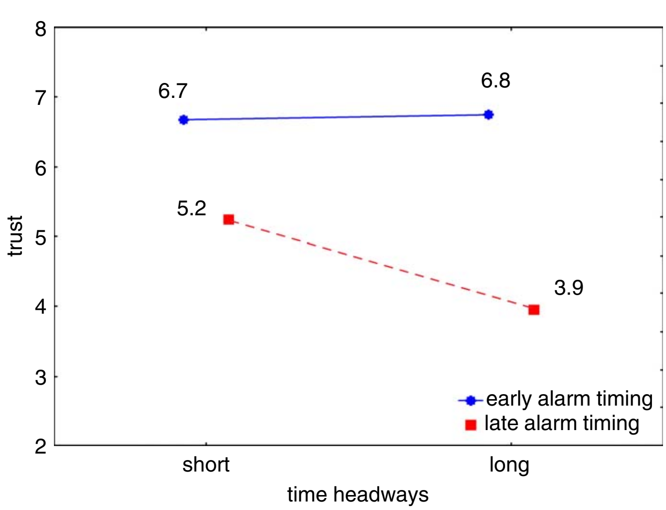

Fig. 6. The effect of time headways on trust according to alarm timing.

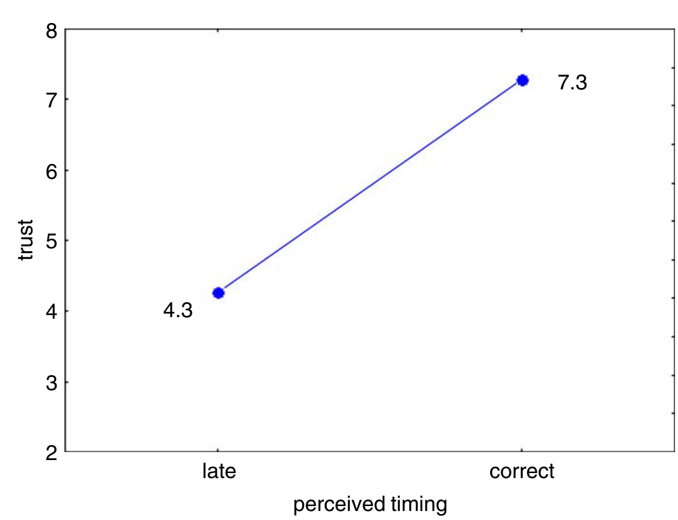

Fig. 7. Trust ratings for perceived late and correct timing.

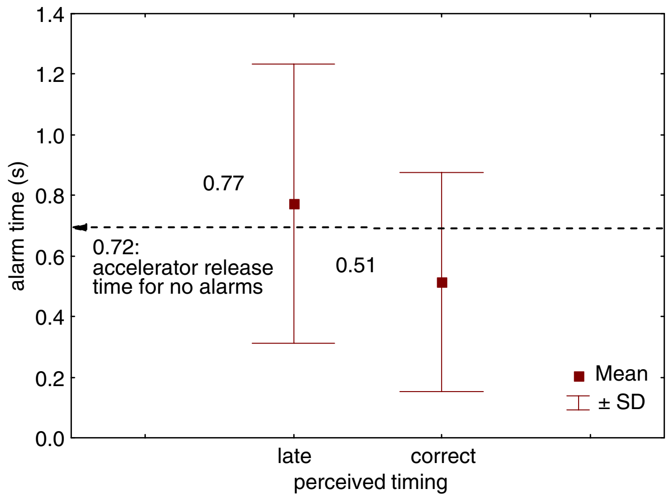

Fig. 8. The relationship between perceived alarm timing and actual alarm time.

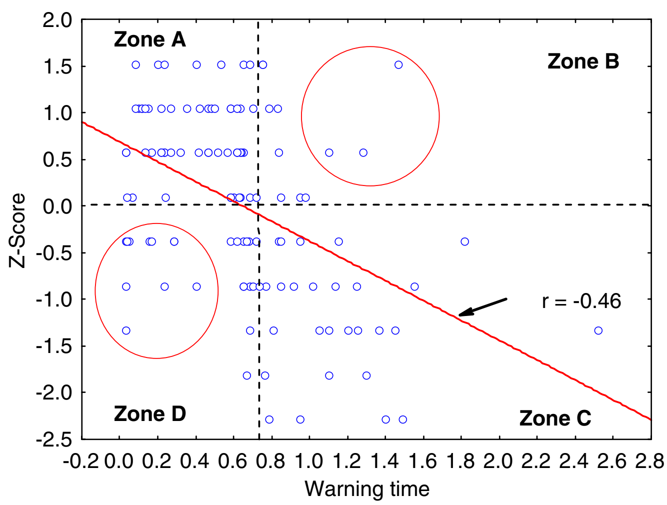

Fig. 9. The relationship between alarm time and trust.

Fig. 8 presents the relationship between perceived alarm timing and actual alarm time. In this figure the dotted line represents the mean value of time for the release of the accelerator when no alarms were given. Interestingly, there is a possibility that if alarms are presented later, compared with driver's own performance in releasing the accelerator without the aid of alarms, then alarms will be viewed as being late.

3.5. The relationship between driver trust and alarm timing

Driver trust and its relation to actual alarm time when alarms were presented after potential collision events happened were considered next. With respect to ratings of trust, a transformation was implemented using Z-scores in order to minimise differences in judgement of trust between individuals and emphasize variation in judgement of trust. First, the values of mean and standard deviation of trust ratings across all trials were calculated. Z-scores were then calculated for each alarm condition in order to provide a standardized rating of each participant's trust. Specifically, the values of Z-scores were found by

$$Z = \{(\text{trust value for each participant and each trial}) - (\text{the mean value of trust for all participants and all trials})\}/(\text{the value of standard deviation for trust ratings for all participant and all trials}).$$

Accordingly, the Z-scores represent how far each participant's trust estimation is different from the centre of distribution of ratings of trust for all participants.

Fig. 9 comprises a scatter plot showing alarm time and trust Z-scores. A negative correlation was found ($R = -0.46$, $p < 0.01$). This result indicates that as alarms are issued later, with respect to potential collision events, then ratings of trust are lower.

Next, data which were not always consistent with the overall trend (i.e. as alarms are provided earlier driver trust is higher) were considered in order to understand their particular characteristics. The horizontal dotted line represents the mean trust value in terms of Z-score (i.e. zero). The vertical dotted line indicates mean value of accelerator release time without the aid of alarms. As can be seen in this figure, all plotted points are assigned to the 4 zones, which are created by the two dotted lines. For the plotted points which are involved in Zones A and C, alarm time and its relation to trust is reasonably understandable as these data can be interpreted using the last analysis of correlation. That is, the earlier the presentation of alarms the higher the trust rating.

However, the data in Zones B and D requires further consideration since some data seem to fall outside the overall trend. In total 10 data and 21 data points were included in Zones B and D, respectively. These data represent relatively late alarms associated with relatively high trust values and alternatively early alarms and low trust ratings. In Zone B 40% of all the data were associated with long time headways. In particular, the three data points which are enclosed by a solid circle seem to lie well outside the trend and these data were all related to driving with long time headways. In Zone D 71% of all the data were associated with short time headways. Also the 10 data points which are enclosed by a dotted circle seem to lie well outside the overall trend and eight data points (80%) were related to driving with short time headways. Table 3 summarised these data in terms of their speed/headway conditions.

The results may suggest that, as an overall trend, as alarms are provided earlier driver trust is higher. On the other hand, in situations where driving demand is relatively low (i.e. when driving with long time headways) driver subjective ratings of trust may exhibit 'tolerance', i.e. trust is relatively high even though alarms are relatively late. Conversely, in situations where driving demand is relatively high, i.e. when driving at high speed with short time

Table 3
Summary of characteristics for error data

Zone B		Zone D	
Driving condition (speed/headway time)	Number of times observed data	Driving condition (speed/headway time)	Number of times of observed data
70/long	2	70/short	5
40/long	1	60/short	2
		40/short	1
		70/long	1
		60/long	1

headways, driver ratings of trust may be 'intolerant', i.e. trust is relatively low even though alarms are relatively early. Clearly, these possibilities are not consistent with the overall trend but do suggest another aspect of the nature of trust; that is trust may be affected by not only alarm timing but also by driving demand.

3.6. Alarm time, driver performance, and trust

In this section, the analyses of alarm time, driver braking performance, trust and their interaction will be integrated in order to provide a human factors perspective in the design of FCWS. Although many factors influence the occurrence of collision events in real traffic only the sudden deceleration of the leading vehicle was used to create potential collision events in the scenarios employed in this study. Accordingly, the following argument can only be applied to these limited collision situations.

It was found that the timing of alarms (i.e. when alarms are activated) was an important contributing factor in determining driver performance, trust and the perception of alarm timing. In this study two predetermined alarm timings were generated by manipulating parameters of the alarm trigger logic (SDA), however the actual alarm timing varied according to driving conditions (three driving speeds and two time headways). Accordingly, the variation in actual alarm time may impact on driver objective and subjective data. In other words, even when the same predetermined alarm timings were used the transition time from a braking event onset to arriving at the warning distance varied in response to the driving conditions. In particular, it was found that time headways may have more potential to influence alarm time than driving speeds. That is, compared with the differences in driving speeds, the differences in time headways may lead to greater variation in alarm time with the same alarm trigger logic.

When driving with short time headways the presentation of early alarms (mean alarm time 0.149 s) reduced driver braking response time and produced a more consistent braking response. Furthermore, trust in these alarms was relatively high. When driving with long time headways, however, early alarms generated by the same trigger logic with the same parameter exhibit a different effect on driver behaviour; there was no improved driver braking response. Interestingly trust in these alarms (mean alarm time 0.557 s) was also relatively high.

The research evidence suggests that improved driver performance with alarms is not always necessary in order to gain relatively high trust ratings and alarm timing may be a more important factor in determining driver trust. As to the overall trend, if alarms are presented earlier trust is higher. It is not surprising that drivers rate early alarms as more trustworthy as these alarms have the potential to give drivers more time in which to implement a braking response to imminent collisions. More importantly, however, some data may exhibit 'tolerance', that is trust is high even when alarms are late when driving with long time headways. Moreover, some data may exhibit 'intolerance', that is trust is low even when alarms are early when driving with short time headways. These results suggest that driver trust may be affected by driving conditions and it is necessary to look at driving demand carefully when considering driver trust. In the near future it is possible that individual differences which may affect driver trust could be incorporated into the design of alarm systems. Such systems could accumulate personal data regarding driving 'style', driver expectation, etc. while driving with collision warning systems, in order to design more personalised driver assistance systems which achieve higher levels of trust and utilisation. The current results may have the potential to contribute to new design strategies for meeting the needs of individual drivers.

As already discussed above, driver perception of alarm time may play an important role in determining driver trust. Accordingly other research questions arise from this argument. What might influence driver perception of alarm timing to explain why alarms may be viewed as early, correct or late? This study suggests a possible answer to this question. The timing of release of the accelerator without alarms, i.e. driver baseline performance (mean value 0.72 s), may have the potential to determine driver perception of alarm timing, i.e. correct timing or late timing. Moreover, whether alarms are actuated before this timing is an important determinant of driver ratings of trust and if alarms are presented after this timing, then trust is impaired.

It is necessary to incorporate these results into the design of alarm timings to enable the development of more effective systems. The finding that the timing of the release of the accelerator can determine perceived late and correct alarms suggests that alarm presentation has to be consistent with driver action to facilitate collision avoidance. Reichart (1993) argued that the behaviour of driver support systems had to be consistent with driver expectation to ensure system efficiency and it can be said that this study found an alarm timing which was consistent with driver expectation resulted in relatively high ratings of trust. It is logical to assume that alarms should be provided as early as possible in order to give drivers more time to respond to critical situations. However, early alarms have been frequently regarded as nuisance alarms or perceived false alarms even when 'true' (for example Janssen et al., 1993) and these alarms may have a negative impact on system efficiency (Hancock and Parasuraman, 1992). The important thing is to balance two conflicting requirements—to deliver a timing that is late enough to minimise the problem of perceived nuisance alarms and early enough to elicit higher ratings of driver trust. This study has provided an idea for optimisation of the design of alarm timing, which reflects driver expectation and may also minimise decreased trust resulting from the mismatch between alarm timing and driver expectation.

Although this study provided important insight into driver response to different alarm timings in several driving conditions and the relation to driver trust, an important limitation of the current study was the drivers' anticipation of the need to brake repeatedly while driving in the simulator. This may result in faster alarm response times than would have been obtained in a real driving environment. It would be necessary to include an experimental condition in which drivers were not primed to anticipate collision events in order to assess absolute values of response time to FCWS in response to alarm timing. Furthermore, this study provided drivers with a very limited alarm presentation. It is likely that driver response to alarms and trust in alarms may vary in response to experience of FCWS and it is necessary to estimate the long-term effects on driver attitudes and trust in order to validate the results observed in this study.

4. Conclusions

This study considered six different driving conditions according to speed and time headway and investigated their effect on driver behaviour and on trust in the presentation of alarms. The results support the following conclusions:

1. When a SDA is employed, differences in driving conditions induce variation in alarm time; the elapsed time from when the leading vehicle begins to brake to when the alarm is presented. This results in different alarm effectiveness and driver perception of alarm timing even if the same pre-determined parameters for triggering alarms are employed.
2. Variation in alarm time mainly derives from differences in time headway rather than from driving speed. Accordingly time headways have a greater influence on the effect of the alarm's presentation on driver performance and perception of alarms.
3. In particular, when driving with long time headways, drivers' adaptation to late alarms may induce a longer response to the brakes compared to the 'no alarm' condition, possibly resulting in impaired driver behaviour.
4. Driver perception of alarm timing reflects driver subjective rating of trust in an alarm relatively well. Accordingly, trust in perceived late alarms is lower than trust in perceived correct alarms. There is a possibility that a threshold between perceived late and correct alarm timings derive from the timing of release of the accelerator for avoiding collisions without the aid of a warning system. That is, when alarms are provided after this time they may be perceived as late alarms.
5. Overall a positive correlation between alarm promptness and driver ratings of trust was found. However, some data indicated another trend describing driver trust and alarm timing and their interactions with driving conditions. Specifically, when driving with long time headways, driver trust in a relatively late timing was not decreased. On the other hand, when driving with short time headways, driver trust in relatively early timing was decreased.

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